

Databases and Web Programming

Week 1 Lab Sheet Solutions

1. Titanic dataset

N/A refers to no specific constraint needed based on the dataset. However, one could define concrete constraints depending on the context and the data types. We will investigate in the next lectures how different values and data types can have different effect on your SQL queries.

Column name	Data type	Constraints	Description	Example value
PassengerId	Integer	Unique. However, there are many datasets that record IDs as combinations of numbers and characters. If that was the case, string would be a better option.	A unique ID assigned to each passenger.	1
Survived	Ideally should be a Boolean (or Integer if you have >2 categories)	0 = Did not survive, 1 = Survived	Indicates whether the passenger survived the disaster.	0
Pclass	Integer	1 = 1st, 2 = 2nd, 3 = 3rd class	The class of the ticket the passenger held.	3
Name	Object (string)	N/A	The full name of the passenger.	"Braund, Mr. Owen Harris"
Sex	Object (string)	Male or Female. This could have also been implemented with Boolean where 1 is male, 0 is female.	The gender of the passenger.	"male"
Age	Integer	Can have missing values	The age of the passenger in years.	22
SibSp	Integer	N/A	The number of siblings or spouses a passenger had aboard.	1
Parch	Integer	N/A	The number of parents or children a passenger had aboard.	0
Ticket	Object (string)	Specific length and characters must be included.	The ticket number for the passenger's fare.	"A/5 21171"
Fare	Float	Should be at least 1.00 euro/pound etc.	The price of the passenger's ticket.	7.25

Cabin	Object (string)	Can have missing values.	The cabin number of the passenger.	"C85"
Embarked	Object (string)	Should have an embarkment location: C, Q, or S	The port from which the passenger embarked (C = Cherbourg, Q = Queenstown, S = Southampton).	"S"

2. Uber rides dataset

N/A refers to no specific constraint needed. However, one could define concrete ones depending on the context and the data types.

Column name	Data type	Constraints	Description	Example value
Date	Object (time)	Not null	The date of the ride booking in YYYY-MM-DD format.	2024-03-23
Time	Object (time)	Not null	The time of the ride booking in HH:MM:SS format.	12:29:38
Booking ID	Object (string)	Not null, Unique	A unique identifier for each ride booking.	"CNR5884300"
Booking Status	Object (string)	Not null	The final status of the ride, such as Completed, Incomplete, or No Driver Found.	"Completed"
Customer ID	Object (string)	Not null, Unique	A unique identifier for each customer.	"CID1982111"
Vehicle Type	Object (string)	Not null, Categorical	The type of vehicle used for the ride.	"Auto"
Pickup Location	Object (string)	Not null	The specific location where the ride began.	"Palam Vihar"
Drop Location	Object (string)	Not null	The specific location where the ride ended.	"Jhilmil"
Avg VTAT	Float	N/A	The average Vehicle Time at Arrival Time. For more complex cases it is important to think of recording the time in minutes, hours etc.	13.4
Avg CTAT	Float	N/A	The average Customer Time at Arrival Time. For more complex cases it is important to think of recording the time in minutes, hours etc.	25.8
Cancelled Rides by Customer	Integer	N/A	Indicates if the ride was cancelled by the customer.	1
Reason for cancelling by Customer	Object (string)	N/A	The reason provided by the customer for cancelling the ride.	"Cancelled by customer"

Cancelled Rides by Driver	Integer	Not null	Indicates if the ride was cancelled by the driver.	1
Driver Cancellation Reason	Object (string)	N/A	The reason provided by the driver for cancelling the ride.	"Vehicle Breakdown"
Incomplete Rides	Integer	N/A	Indicates if the ride was incomplete.	1
Incomplete Rides Reason	Object (string)	N/A	The reason for the incomplete ride.	"Vehicle Breakdown"
Booking Value	Float	N/A	The total value of the booking in currency.	627
Ride Distance	Float	Not null	The distance of the ride in kilometres. Even if the drive was cancelled it should be "0.00"	13.58
Driver Ratings	Float	New drivers might not have ratings yet hence Null values. Better would be to add constraint with "0".	The ratings of a driver (possibly average from all rides).	4.9
Customer Rating	Float	New customers might not have ratings yet hence Null values. Better would be to add constraint with "0".	The ratings of a customer (possibly average from all rides).	4.1
Payment method	Object (string)	Payment can be Null only if the ride has been cancelled. A better way to design the dataset would be to add "0" instead of null.	Method used for payment.	"Cash"

The analytical competencies you developed through this exercise—namely, understanding data structure, semantics (i.e. meaning of data in a concrete context) and defining constraints—are **foundational and broadly applicable**. They are crucial for:

- **Database Definition:** Ensuring the accurate and robust modelling and implementation of relational and graph databases.
- **Advanced Data Tasks:** Preparing and validating datasets for subsequent use in **machine learning (ML)** and **artificial intelligence (AI)** applications.

<https://www.kaggle.com/datasets/yashdevladdha/uber-ride-analytics-dashboard/data>

<https://www.kaggle.com/datasets/yasserh/titanic-dataset>

A thorough **understanding of the underlying data** is the prerequisite for its correct and efficient **structuring within a database environment**.